

The Effect of Reward Function Design in RL-Based BTC Grid Trading: Pareto Frontier, Winner Reversal, and Out-of-Distribution Robustness of Prospect-Theoretic Reward

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Abstract

We quantitatively study the effect of *reward function design* on PPO-based reinforcement-learning policies for ATR-proportional grid trading on the BTC/USDT 1-hour market. Four reward variants—symmetric (**sym**), asymmetric ($\beta = 3.42$, **asym**), Differential Sharpe Ratio ($\eta = 1/28$ h, **dsr**; [Moody and Saffell 2001](#)), and prospect-theoretic ($\alpha = 0.68$, $\lambda = 3.30$, **pt**; [Kahneman and Tversky 1979](#))—are each trained with $10 \text{ seeds} \times 1\text{M}$ steps and evaluated on three environments: (i) single-split validation (Val 2021–2023), (ii) 6-fold combinatorial purged cross-validation [[López de Prado, 2018](#)] 15 paths, and (iii) the unsealed out-of-sample Test set (2024–2026, BTC \$42K $\rightarrow$ \$76K bull regime).

We report five key findings. **(1)** All four variants exceed the ATR baseline by +11–24% on Val, yet partition into two clusters $\{\text{sym}, \text{dsr}\}$ (aggressive) vs. $\{\text{asym}, \text{pt}\}$ (conservative); within-cluster Cohen’s $|d| < 0.3$ versus across-cluster $|d| > 0.8$. **(2)** Mechanism: reward loss asymmetry directly determines trade frequency and holding time (policy distance ratio 2.22 $\times$), and the sliding-window memory structure of DSR results in 2–6 $\times$ higher hold rate than other variants. **(3)** Winner reverses across evaluation environments: Val=**sym**(1.87) $\rightarrow$ CPCV=**dsr**(1.41, $p < 0.001$) $\rightarrow$ Test=**pt**(0.34–0.37, consistent across two model sources, $p < 0.002$). **(4)** The OOS robustness of **pt** arises from a short-holding mechanism: loss aversion $\lambda = 3.30$ combined with concave gain $\alpha = 0.68$ trains policies to exit within mean 1.4 h (max 6 h), thereby avoiding the sell-side timing risk of the unseen bull regime. In contrast, **dsr** learns long holding (mean 4.58 h, max 169 h = 7 days) and records the worst OOS performance—demonstrating that the same reward formulation underlies both in-sample advantage and OOS failure as two sides of the same coin. **(5)** The Val–Test distribution shift is highly significant (KS $p < 10^{-10}$, –27% variance), while policies’ actions remain nearly invariant between Val and Test ($\Delta \leq 5\%$), indicating that the regime shift, not the policy, drives OOS differences.

Our contribution is twofold. First, we demonstrate quantitatively that reward design

exerts its effect through a *trade-off dimension in risk profile* rather than a single Sharpe “alpha source.” Second, we provide the first quantitative evidence that Kahneman-Tversky prospect theory confers out-of-distribution safety on RL trading policies. Code and data: <https://github.com/cosmicpotato2047/capstone-rl-trading>.

Keywords: Reinforcement Learning, Grid Trading, PPO, Reward Design, Prospect Theory, Differential Sharpe Ratio, Combinatorial Purged CV, Out-of-Distribution Generalization, Cryptocurrency

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1 Introduction

Cryptocurrency markets exhibit micro-oscillations in hourly price that are largely orthogonal to long-term trends, motivating *grid trading* strategies that harvest revenue from volatility itself rather than from directional prediction. Grid trading is a simplified form of market making [Avellaneda and Stoikov, 2008], ranging from fixed-grid rules to volatility-adaptive variants that scale grid width by the Average True Range (ATR) [Wilder, 1978]. This work studies the latter: an ATR-proportional grid system whose grid width and limit-order placement are determined by a reinforcement-learning policy.

1.1 Motivation and Research Question

An earlier phase of this project hypothesized that PPO RL would discover dynamic volatility adaptation and outperform a fixed-grid baseline. Post-hoc behavioral analysis, however, revealed that the trained policy converged to a near-constant action [1.0, 0.0] (§4), and that the Bayesian-optimized ATR coefficient itself was the principal source of performance. This negative finding suggested that the symmetric-reward + ATR-proportional combination structurally absorbs the policy’s degrees of freedom, motivating the new research question of this paper:

In BTC grid trading, how does the design of the reward function affect the behavioral pattern and generalization performance of RL policies? In particular, do loss-asymmetric (asymmetric, prospect-theoretic) or risk-adjusted-return (DSR) reward formulations yield meaningful differences in single-metric advantage or out-of-sample robustness?

1.2 Contributions

This work makes the following four contributions.

1. **A statistically rigorous comparison framework for reward variants.** Four reward functions (`sym`, `asym`, `dsr`, `pt`) are trained in an identical environment with $10 \text{ seeds} \times 1\text{M}$ steps, and pairwise statistical tests (Cohen’s d , bootstrap $P(A > B)$, IQM, 5% CVaR) are applied following Henderson et al. [2018]’s reproducibility guidelines.
2. **Scenario D — Pareto-frontier discovery.** Rather than a single “best variant” winner, the four variants statistically partition into two clusters $\{\text{sym}, \text{dsr}\}$ vs. $\{\text{asym}, \text{pt}\}$ forming a Pareto-like frontier on the Sharpe–MDD plane (§5). This outcome falls into none of the pre-registered scenario branches (A optimistic / B neutral / C pessimistic).
3. **Causal chain: reward $\rightarrow$ behavior $\rightarrow$ outcome.** Policy-level differences across variants are quantified on 1.04M step trajectories via a policy-distance matrix (within-cluster 0.129 vs. across-cluster 0.286), regime- specific trade/hold statistics, and action distribution comparison, confirming the mechanism reward loss asymmetry $\rightarrow$ reduced trade frequency $\rightarrow$ conservative cluster (§6).
4. **Winner reversal across evaluation environments and OOS robustness of prospect theory.** The winner reverses completely across single-split Val (`sym` 1st) $\rightarrow$ CPCV multi-split

(**dsr** 1st) $\rightarrow$ the unsealed Test (**pt** 1st, consistent across two sources $p < 0.002$) (§7). The OOS robustness of **pt** is mechanistically attributed via trajectory analysis to its loss aversion $\lambda = 3.30$ and concave utility $\alpha = 0.68$ training policies to exit within mean 1.4h, thereby avoiding the sell-side timing risk of the unseen bull regime. This is, to our knowledge, the first quantitative demonstration that the Kahneman-Tversky prospect theory [Kahneman and Tversky, 1979] confers OOS safety on RL trading policies.

1.3 Paper Structure

Section 2 surveys related work on RL trading, reward design theory, and evaluation methodology. Section 3 details the environment design (MDP, ATR formula, four reward variants) and PPO training. Section 4 briefly reproduces the Phase 1 negative finding (“RL = Fixed [1.0, 0.0]”) as motivation. Section 5 reports the two-cluster Pareto-frontier result on Val (Scenario D), while Section 6 quantifies the mechanism. Section 7 presents the robustness trio: slippage (§7.1), CPCV multi-split (§7.2), and the central contribution of this paper—the Test OOS evaluation and the mechanistic basis of prospect theory’s robustness (§7.3). Section 8 discusses distribution shift quantification and limitations, and Section 9 concludes.

2 Related Work

This paper sits at the intersection of four areas: (i) market making and grid trading, (ii) reinforcement-learning-based trading, (iii) reward-design theory, and (iv) evaluation methodology for trading policies. This section surveys the key prior work in each area and clarifies where our contribution lies.

2.1 Market Making and Grid Trading

The academic ancestor of grid trading is the market-making model of Avellaneda and Stoikov [2008]. Their analysis derives a closed-form optimal policy for a risk-averse market maker who faces inventory risk and must place *asymmetric* bid/ask quotes around the mid-price. Two key insights are that (a) the bid–ask spread is a function of volatility and remaining time horizon, and (b) inventory skew shifts the quote midpoint asymmetrically. Grid trading is a simplified practical variant of this model in which orders are placed on a pre-defined ladder of price levels in order to monetize micro-volatility.

The standard mechanism for adapting grid widths to market volatility is the Average True Range [Wilder, 1978] (ATR; 14-period moving average is the default). ATR is a smoothed average of the true price range and provides a simple, robust rule that automatically widens the grid in volatility-cluster regimes. ATR-proportional grids of the form $\Delta = c \cdot \text{ATR}$ (with proportional coefficient c) are widely used in practice and serve as the baseline of this paper.

Most prior academic work in this area is limited to the analysis of *rule-based* policies, with relatively few attempts to learn the grid-width decision from data. This paper adopts a structure in which the proportional coefficient and the number of grid splits in the ATR formula are dynamically determined by an RL policy, while quantitatively analyzing how the design of

the reward function influences the resulting learned policy’s behavior—a combination, to our knowledge, that has not been explored.

2.2 Reinforcement Learning for Trading

The standard DRL baseline for cryptocurrency trading is the PPO/A2C/DQN comparison framework of Zhang et al. [2020], which feeds price time series directly into a PPO policy and shows consistent improvements over simple buy-and-hold (B&H) and moving-average crossover strategies. The ACM survey of Sun et al. [2023] and the more recent review of Liu [2025] both identify two open problems in the area: (a) *interpretability of policy behavior* and (b) *generalization across market regimes*. This paper addresses both directly: §6 quantifies the causal chain from reward form to behavioral difference, and §7 documents out-of-sample (OOS) generalization phenomena that are not captured by single-split validation alone.

There are four directly preceding works. Liu et al. [2020] feeds the output of an LSTM price predictor into the state of a PPO trading policy in a two-stage pipeline. The limitation of this approach is that prediction errors propagate directly into trading decisions; this paper bypasses the limitation by *eliminating* the price-prediction stage and using a volatility-adaptive grid structure instead. Yasin and Gill [2024] compares DQN/PPO/A2C with 20 technical indicators, but the action space is discrete buy/sell with no hold action. This paper adopts a continuous two-dimensional action (aggressiveness and profit target) that supports fine-grained grid control. Pham et al. [2025] uses DQN to select one of five predefined strategies as a meta-strategy, but the strategies themselves are not learned. Bandarupalli [2025] augments PPO with a volatility penalty and a transaction-cost term, but stays at the level of hyperparameter tuning within a single reward variant and does not compare across reward *forms*, which is the focus of this paper.

A particularly relevant prior work is Gort et al. [2022], which empirically shows that *single-split train/test results do not guarantee OOS generalization* in cryptocurrency DRL trading, and recommends CPCV (Combinatorial Purged Cross-Validation) and DSR (Deflated Sharpe Ratio). This paper adopts those recommendations, using Val (2021–2023), CPCV (6-fold inside the Train partition), and Test (2024–unsealed) environments concurrently. Our *winner reversal* finding (Val **sym** $\rightarrow$ CPCV **dsr** $\rightarrow$ Test **pt**) is a quantitative confirmation that Gort et al.’s concern reappears in the dimension of reward-variant selection.

In one sentence, our position is: whereas prior DRL trading literature focuses on comparing algorithms and state designs under a *single* reward function, this paper performs a controlled comparison of *four reward functions* on top of the same algorithm (PPO) and the same environment, and separately measures their effects on in-sample / multi-split / OOS environments.

2.3 Reward-Design Theory

The four reward variants compared in this paper draw from three theoretical lineages in reinforcement learning and behavioral economics. (i) Symmetric (**sym**) is the standard PnL-return reward and serves as our baseline. (ii) Asymmetric (**asym**) places a positive weight $\beta > 1$ on losses, a simple first-order approximation of a risk-averse utility. (iii) Differential Sharpe Ratio

(**dsr**) is the formulation of [Moody and Saffell \[2001\]](#). (iv) Prospect-theoretic reward (**pt**) directly uses the utility function of [Kahneman and Tversky \[1979\]](#) and [Tversky and Kahneman \[1992\]](#), $v(x) = \text{sgn}(x)|x|^\alpha$ for $x \geq 0$ and $-\lambda|x|^\alpha$ for $x < 0$, as a per-step reward.

The DSR formulation of [Moody and Saffell \[2001\]](#) uses the *differential change of the online Sharpe-ratio estimator* as the step reward, tracking first and second moments via exponentially-weighted moving averages (EWMA, decay rate η). This has two implications for RL training: (a) it directly optimizes a risk-adjusted return rather than raw PnL, and (b) the sliding-window memory imposes a *multi-step temporal dependency* on the policy. As we show in §6, this memory structure is what causes the **dsr** policy to learn 2–6× longer holding times than the other variants—a property that becomes the in-sample advantage of §7 and the OOS failure of §7.3 as two sides of the same causal coin.

Prospect theory [[Kahneman and Tversky, 1979](#)] is a foundational model of human decision making under risk, with two characteristic parameters: (a) loss aversion $\lambda \approx 2.25$ (standard value from [Tversky and Kahneman 1992](#)) and (b) the concavity exponent $\alpha \approx 0.88$. While the theory has been widely used to describe human trader behavior (e.g., the disposition effect of clinging to losses and realizing gains too early), *direct use of prospect-theoretic utility as an RL reward* is rare in the literature. To our knowledge, this paper is the first to directly implement the [Kahneman and Tversky \[1979\]](#) utility as an RL reward (with $\alpha = 0.68$, $\lambda = 3.30$, tuned by Optuna) and quantitatively establish the OOS robustness that the resulting policy exhibits.

[Ng et al. \[1999\]](#)’s policy-invariance theorem of potential-based reward shaping states that “a monotone transformation of the reward signal does not alter the policy learned by RL.” At first glance, this paper’s comparison of four reward variants may appear to conflict with the theorem, but it does not: the **sym** $\rightarrow$ **asym**/**pt** transformations are *non-potential-based* (they apply an asymmetric scale on the negative half-line), while **dsr** is not a simple per-step reward transformation but a *state-extended* reward (the sliding-window moments effectively become part of the state). Our reward-variant comparison thus lies outside the scope of the Ng et al. theorem, and indeed we confirm in §6 that policy behavior differs across variants in a statistically significant manner.

2.4 Evaluation Methodology

The two principal pitfalls in evaluating RL trading policies are (a) backtest overfitting and (b) seed dependence of results. [Henderson et al. \[2018\]](#) empirically demonstrates that seed variability alone can produce large differences for the same algorithm, and proposes 5–10 seed statistical comparisons, IQM (Interquartile Mean), and bootstrap confidence intervals as standard recommendations. This paper uses 10 seeds per variant (Val/exp033) and 100 models (Test), and applies pairwise Cohen’s d and bootstrap $P(A > B)$ to all comparisons.

[López de Prado \[2018\]](#) (Ch. 8) proposes *Combinatorial Purged Cross-Validation* (CPCV), which avoids train/test leakage in time-series data while leveraging $\binom{N}{k}$ split combinations to strengthen the generalization guarantees of single-split results. We adopt 6-fold CPCV (15 paths) in §7.2.

Bailey and López de Prado [2014]’s *Deflated Sharpe Ratio* (DSR^*) is a statistical-significance test on the Sharpe ratio that corrects for the number of backtest trials and for the skewness/kurtosis of the underlying distribution. We apply Bonferroni 4-way correction together with DSR^* testing to the multiple comparisons of CPCV results (§7.2), and confirm that all four reward variants are significant relative to the ATR baseline at $p < 0.001$.

The methodological distinctiveness of this paper is the *simultaneous use of three evaluation environments* (Val / CPCV / Test) that allowed us to discover the winner reversal. Studies relying solely on single-split or solely on CPCV cannot expose OOS generalization failure, and studies relying solely on OOS cannot expose the in-sample conservatism of **pt**. The simultaneous application of three environments is, in our view, a methodological contribution: it generalizes the multi-evaluation recommendations of Henderson et al. [2018] and Gort et al. [2022] to the single dimension of reward-variant comparison.

3 Method

[TBD. Subsections: §3.1 MDP (State 7D, Action 2D, ATR-proportional formula); §3.2 trading-env mechanics (cycles, fee, slippage); §3.3 four reward variants + Optuna tuning (exp032a); §3.4 PPO stabilization (target_kl, ent annealing); §3.5 evaluation protocol (Val / CPCV / Test).]

4 Phase 1: Policy Saturation under Symmetric Reward

[TBD. Reproduces the “ $RL = \text{Fixed } [1.0, 0.0]$ ” finding from exp020–022 of the earlier phase. Motivates reward formulation in this paper.]

5 Pareto Frontier and Scenario D

[TBD. exp032b results: 4 variants $\times$ 10 seeds, pairwise statistics, Pareto scatter, two-cluster discovery, hypothesis H1–H3 evaluation, Scenario D confirmation.]

6 Mechanism Quantification

[TBD. exp032c: policy distance matrix, behavior per regime, counterfactual action map, action distribution; causal chain reward $\rightarrow$ trade frequency $\rightarrow$ cluster.]

7 Robustness Analysis

7.1 Slippage Robustness

[TBD. exp033 results with slippage 0.02%. Cluster preservation ratio 2.19 $\times$; ATR-with-slippage fair-comparison results (Phase 16a).]

7.2 Multi-Split Evaluation via CPCV

[TBD. *exp034*: 6-fold CPCV 15 paths. DSR $p < 0.001$ (Bonferroni 4-way correction); **dsr** reversal to 1st (1.413, 5% CVaR 0.890); cluster preservation ratio 3.55×.]

7.3 Out-of-Sample Evaluation and Prospect Theory’s Robustness

[TBD. Central chapter. *exp035*: 100 RL models + ATR baseline + Buy-and-Hold on Test. **pt** 1st on Test (both sources $p < 0.002$); **dsr** last on Test. Phase 16d mechanism: hold-duration distribution (DSR 7-day max vs. **pt** 6h max); causal account of **pt**’s OOS robustness.]

8 Discussion

[TBD. Subsections: §8.1 Val–Test distribution-shift quantification (KS $p < 10^{-10}$, −27% variance); §8.2 honest acknowledgment of environment dependence in *exp027_rl* prior evidence (Env-v3 **asym** 1.955); §8.3 limits of single-metric winner; §8.4 limitations: BTC-only, single regime in Test, single slippage level.]

9 Conclusion

[TBD. Four main messages: (1) Pareto-frontier trade-off; (2) winner reversal across evaluation environments; (3) OOS robustness of Prospect Theory; (4) in-sample advantage $\neq$ OOS robustness. Future work: multi-asset, additional regimes, DR, meta-RL.]

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